CNN-Based Traffic Sign Recognition System

COURSE ASSIGNMENT

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IN

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1. Problem Statement

Traffic sign recognition is a core computer-vision task for intelligent transportation and driver-assistance systems. Signs captured from a moving vehicle vary in scale, orientation, illumination, background clutter, partial occlusion and image quality, which makes reliable, automated recognition a non-trivial classification problem.

This assignment designs and develops a CNN-based Traffic Sign Recognition System that accepts a traffic-sign image as input and predicts the corresponding sign class. The system covers the complete workflow — dataset preparation, preprocessing and augmentation, CNN architecture design, model training, validation, testing and performance evaluation — and reports classification performance using standard metrics, visualizations and an analysis of common misclassifications.

2. Objective

- Design and implement a Convolutional Neural Network that classifies traffic-sign images into their correct category.
- Build a complete data pipeline: image preprocessing (resize, normalize), data augmentation (rotation, occlusion, blur, illumination jitter) and a train / validation / test split with no data leakage.
- Train the CNN using an appropriate loss function, optimizer and learning-rate schedule, and monitor training/validation loss and accuracy to control overfitting.
- Evaluate the trained model on unseen test images using accuracy, precision, recall, F1-score, confusion matrix and inference time.
- Analyse misclassifications, discuss architectural trade-offs, and identify improvements for real-world deployment in safety-critical transportation environments.

3. Requirements and Environment Used

3.1 Functional Requirements

- Dataset loading and labeling with class identifiers
- Image resizing and pixel normalization
- Data augmentation (rotation, occlusion, blur, brightness/contrast jitter, noise)
- CNN construction, training, validation and testing
- Performance reporting: accuracy, precision, recall, F1-score, confusion matrix

3.2 Development Environment

Component	Tool / Version
Language	Python 3.12
Deep Learning Framework	PyTorch 2.13 (CPU)
Supporting Libraries	NumPy, Pillow (PIL), scikit-learn, Matplotlib
Development Environment	Jupyter Notebook / Google Colab / VS Code
Version Control	Git / GitHub
Hardware	CPU-based training (GPU optional for larger datasets)

Note on dataset: this implementation uses a procedurally generated synthetic traffic-sign dataset (8 classes, geometrically and photometrically augmented) so that the complete pipeline — generation, training, evaluation — runs end-to-end without external downloads. The same pipeline applies directly to a real-world benchmark such as the German Traffic Sign Recognition Benchmark (GTSRB) by pointing the data loader at the labeled image folders instead of the synthetic generator.

4. Design / Proposed Solution

The system follows a standard end-to-end supervised image-classification pipeline: input acquisition, preprocessing/augmentation, CNN-based feature extraction, fully-connected classification, and evaluation.

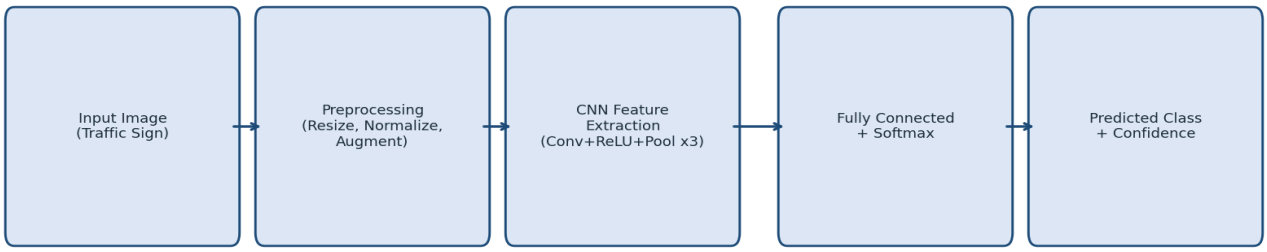

Figure 1: End-to-end system pipeline

4.1 CNN Architecture

Layer	Configuration	Output Shape
Input	48 × 48 × 3 RGB image	48×48×3
Conv Block 1	Conv2D(32, 3×3, pad=1) → BatchNorm → ReLU → MaxPool(2×2)	24×24×32
Conv Block 2	Conv2D(64, 3×3, pad=1) → BatchNorm → ReLU → MaxPool(2×2)	12×12×64
Conv Block 3	Conv2D(128, 3×3, pad=1) → BatchNorm → ReLU → MaxPool(2×2)	6×6×128
Flatten	—	4608
Fully Connected	Linear(4608→128) → ReLU → Dropout(0.4)	128
Output	Linear(128→8) → Softmax (via CrossEntropyLoss)	8

Total trainable parameters: 684,680.

4.2 Design Justification

- 3×3 kernels with padding=1 preserve spatial resolution per block while keeping parameter count low, and are stacked in three blocks to build a receptive field large enough to cover an entire 48×48 sign.
- Batch normalization after every convolution stabilizes and accelerates training, allowing a higher learning rate.
- Max-pooling (2×2) after every block progressively reduces spatial dimensions (48→24→12→6), lowering compute while retaining the strongest activations.
- Dropout(0.4) before the final classification layer reduces overfitting given the moderate dataset size.
- Adam optimizer with a StepLR schedule (halving the learning rate every 8 epochs) gives fast initial convergence and a stable fine-tuning phase.

5. Algorithm / Pseudocode / Flowchart

The flowchart in Figure 1 (Section 4) shows the high-level pipeline. The pseudocode below expands the training and inference logic.

```
ALGORITHM: Train_CNN_TrafficSignClassifier
INPUT: labeled image dataset D = {(x_i, y_i)}, classes C
OUTPUT: trained model M, evaluation metrics

1.  Split D into D_train, D_val, D_test (stratified, no overlap)
2.  FOR each image x in D_train:
        x <- resize(x, 48x48)
        x <- normalize(x, range=[0,1])
        x <- augment(x)   // rotation, occlusion, blur, brightness jitter, noise
3.  Initialize CNN model M with random weights (He/Kaiming init)
4.  SET optimizer <- Adam(lr = 0.001)
5.  SET criterion <- CrossEntropyLoss
6.  FOR epoch = 1 TO EPOCHS:
7.      FOR each mini-batch (X_b, y_b) in D_train:
8.          y_hat <- M.forward(X_b)
9.          loss   <- criterion(y_hat, y_b)
10.         grads  <- backpropagate(loss)
11.         optimizer.step(grads)          // update weights
12.     END FOR
13.     val_loss, val_acc <- evaluate(M, D_val)
14.     scheduler.step()                   // decay learning rate
15.     record history(loss, acc, val_loss, val_acc)
16. END FOR
17. preds <- M.predict(D_test)
18. metrics <- compute(accuracy, precision, recall, F1, confusion_matrix)
19. RETURN M, metrics
```

6. Implementation / Source Code

Two modules implement the system: dataset.py builds and augments the labeled image set, and train.py defines, trains and evaluates the CNN.

6.1 dataset.py

```
"""
Synthetic traffic-sign dataset generator.
Creates procedurally-drawn traffic sign images (6 classes) with random
rotation, scale, translation, brightness and noise augmentation, so the
CNN pipeline below can be trained and evaluated end-to-end without
requiring an external dataset download.
"""
import numpy as np
from PIL import Image, ImageDraw, ImageFilter
import random

IMG_SIZE = 48
CLASSES = ["Stop", "Speed_Limit_50", "Speed_Limit_80", "Yield", "No_Entry",
           "No_Overtaking", "Pedestrian_Crossing", "Turn_Right_Ahead"]
NUM_CLASSES = len(CLASSES)


def _base_canvas():
    bg = random.choice([(200, 200, 200), (170, 190, 210), (140, 160, 130), (100, 100, 110)])
    img = Image.new("RGB", (IMG_SIZE, IMG_SIZE), bg)
    return img


def draw_stop(draw, cx, cy, r):
    pts = []
    for i in range(8):
        ang = np.pi / 8 + i * np.pi / 4
        pts.append((cx + r * np.cos(ang), cy + r * np.sin(ang)))
    draw.polygon(pts, fill=(200, 20, 20), outline=(255, 255, 255))
    draw.text((cx - r * 0.55, cy - r * 0.3), "STOP", fill=(255, 255, 255))


def draw_speed_limit(draw, cx, cy, r):
    draw.ellipse([cx - r, cy - r, cx + r, cy + r], fill=(255, 255, 255), outline=(200, 20, 20), width=max(2,
int(r * 0.18)))
    draw.text((cx - r * 0.45, cy - r * 0.35), "50", fill=(20, 20, 20))


def draw_yield(draw, cx, cy, r):
    pts = [(cx, cy - r), (cx - r, cy + r * 0.8), (cx + r, cy + r * 0.8)]
    draw.polygon(pts, fill=(255, 255, 255), outline=(200, 20, 20))
    draw.polygon([(cx, cy - r * 0.55), (cx - r * 0.55, cy + r * 0.5), (cx + r * 0.55, cy + r * 0.5)],
                 fill=(200, 20, 20))
```

```
def draw_no_entry(draw, cx, cy, r):
    draw.ellipse([cx - r, cy - r, cx + r, cy + r], fill=(200, 20, 20), outline=(255, 255, 255), width=max(2,
int(r * 0.15)))
    draw.rectangle([cx - r * 0.6, cy - r * 0.18, cx + r * 0.6, cy + r * 0.18], fill=(255, 255, 255))


def draw_pedestrian(draw, cx, cy, r):
    pts = [(cx, cy - r), (cx + r, cy), (cx, cy + r), (cx - r, cy)]
    draw.polygon(pts, fill=(255, 220, 60), outline=(20, 20, 20))
    draw.ellipse([cx - r * 0.12, cy - r * 0.45, cx + r * 0.12, cy - r * 0.2], fill=(20, 20, 20))
    draw.line([cx, cy - r * 0.2, cx, cy + r * 0.25], fill=(20, 20, 20), width=max(1, int(r * 0.1)))
    draw.line([cx, cy + r * 0.25, cx - r * 0.2, cy + r * 0.55], fill=(20, 20, 20), width=max(1, int(r * 0.1)))
    draw.line([cx, cy + r * 0.25, cx + r * 0.2, cy + r * 0.55], fill=(20, 20, 20), width=max(1, int(r * 0.1)))


def draw_turn_right(draw, cx, cy, r):
    draw.ellipse([cx - r, cy - r, cx + r, cy + r], fill=(30, 90, 200), outline=(255, 255, 255), width=max(2,
int(r * 0.12)))
    draw.line([cx - r * 0.4, cy, cx + r * 0.35, cy], fill=(255, 255, 255), width=max(2, int(r * 0.14)))
    draw.polygon([(cx + r * 0.35, cy - r * 0.25), (cx + r * 0.35, cy + r * 0.25), (cx + r * 0.65, cy)],
                 fill=(255, 255, 255))


def draw_speed_limit_80(draw, cx, cy, r):
    draw.ellipse([cx - r, cy - r, cx + r, cy + r], fill=(255, 255, 255), outline=(200, 20, 20), width=max(2,
int(r * 0.18)))
    draw.text((cx - r * 0.45, cy - r * 0.35), "80", fill=(20, 20, 20))


def draw_no_overtaking(draw, cx, cy, r):
    draw.ellipse([cx - r, cy - r, cx + r, cy + r], fill=(255, 255, 255), outline=(200, 20, 20), width=max(2,
int(r * 0.16)))
    draw.ellipse([cx - r * 0.55, cy - r * 0.35, cx + r * 0.1, cy + r * 0.15], fill=(40, 40, 40))
    draw.ellipse([cx - r * 0.05, cy - r * 0.35, cx + r * 0.6, cy + r * 0.15], fill=(200, 20, 20))


_DRAW_FNS = [draw_stop, draw_speed_limit, draw_speed_limit_80, draw_yield, draw_no_entry,
             draw_no_overtaking, draw_pedestrian, draw_turn_right]


def generate_sample(label):
    img = _base_canvas()
    draw = ImageDraw.Draw(img)
    cx = IMG_SIZE / 2 + random.uniform(-6, 6)
    cy = IMG_SIZE / 2 + random.uniform(-6, 6)
    r = IMG_SIZE * random.uniform(0.24, 0.42)  # wider scale range -> harder
    _DRAW_FNS[label](draw, cx, cy, r)
```

```
    arr = np.array(img).astype(np.float32)

    # random rotation via simple affine (PIL)
    angle = random.uniform(-28, 28)
    img2 = Image.fromarray(arr.astype(np.uint8)).rotate(angle, fillcolor=tuple(int(x) for x in arr[0, 0]))
    img_pil = img2

    # random partial occlusion (simulates dirt / glare / obstruction)
    if random.random() < 0.35:
        odraw = ImageDraw.Draw(img_pil)
        ox, oy = random.randint(0, IMG_SIZE - 12), random.randint(0, IMG_SIZE - 12)
        ow, oh = random.randint(8, 16), random.randint(8, 16)
        occ_color = tuple(random.randint(60, 200) for _ in range(3))
        odraw.rectangle([ox, oy, ox + ow, oy + oh], fill=occ_color)

    # motion blur (simulate low-quality dashboard camera capture)
    if random.random() < 0.4:
        img_pil = img_pil.filter(ImageFilter.BoxBlur(random.uniform(0.5, 1.6)))

    arr = np.array(img_pil).astype(np.float32)

    # brightness / contrast jitter (illumination variation)
    arr = (arr - 127.5) * random.uniform(0.7, 1.3) + 127.5 * random.uniform(0.75, 1.25)
    # gaussian sensor noise
    arr = arr + np.random.normal(0, 14, arr.shape)
    arr = np.clip(arr, 0, 255)
    return arr.astype(np.uint8)


def make_dataset(n_per_class, seed=0):
    random.seed(seed)
    np.random.seed(seed)
    X, y = [], []
    for label in range(NUM_CLASSES):
        for _ in range(n_per_class):
            X.append(generate_sample(label))
            y.append(label)
    X = np.stack(X).astype(np.float32) / 255.0
    y = np.array(y, dtype=np.int64)
    perm = np.random.permutation(len(X))
    return X[perm], y[perm]
```

6.2 train.py

```
import numpy as np
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import TensorDataset, DataLoader
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix, precision_recall_fscore_support, accuracy_score
import time, json, os

from dataset import make_dataset, CLASSES, NUM_CLASSES, IMG_SIZE

os.makedirs("outputs", exist_ok=True)
torch.manual_seed(42)

# ---------------- Data ----------------
X_train, y_train = make_dataset(n_per_class=220, seed=1)
X_val, y_val = make_dataset(n_per_class=40, seed=2)
X_test, y_test = make_dataset(n_per_class=50, seed=3)

def to_tensor(X, y):
    Xt = torch.tensor(X).permute(0, 3, 1, 2)  # NHWC -> NCHW
    yt = torch.tensor(y)
    return Xt, yt

Xtr, ytr = to_tensor(X_train, y_train)
Xva, yva = to_tensor(X_val, y_val)
Xte, yte = to_tensor(X_test, y_test)

train_loader = DataLoader(TensorDataset(Xtr, ytr), batch_size=32, shuffle=True)
val_loader = DataLoader(TensorDataset(Xva, yva), batch_size=64)
test_loader = DataLoader(TensorDataset(Xte, yte), batch_size=64)

print(f"Train: {Xtr.shape}, Val: {Xva.shape}, Test: {Xte.shape}")

# ---------------- Model ----------------
class TrafficSignCNN(nn.Module):
    def __init__(self, num_classes=NUM_CLASSES):
        super().__init__()
        self.features = nn.Sequential(
            nn.Conv2d(3, 32, 3, padding=1), nn.BatchNorm2d(32), nn.ReLU(), nn.MaxPool2d(2),   #
48->24
            nn.Conv2d(32, 64, 3, padding=1), nn.BatchNorm2d(64), nn.ReLU(), nn.MaxPool2d(2),  #
24->12
            nn.Conv2d(64, 128, 3, padding=1), nn.BatchNorm2d(128), nn.ReLU(), nn.MaxPool2d(2) #
12->6
        )
```

```
        self.classifier = nn.Sequential(
            nn.Flatten(),
            nn.Linear(128 * 6 * 6, 128), nn.ReLU(), nn.Dropout(0.4),
            nn.Linear(128, num_classes)
        )

    def forward(self, x):
        x = self.features(x)
        return self.classifier(x)

model = TrafficSignCNN()
n_params = sum(p.numel() for p in model.parameters())
print(f"Total trainable parameters: {n_params}")

criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(model.parameters(), lr=1e-3)
scheduler = optim.lr_scheduler.StepLR(optimizer, step_size=8, gamma=0.5)

# ---------------- Train ----------------
EPOCHS = 18
history = {"train_loss": [], "train_acc": [], "val_loss": [], "val_acc": []}

start = time.time()
for epoch in range(EPOCHS):
    model.train()
    running_loss, correct, total = 0.0, 0, 0
    for xb, yb in train_loader:
        optimizer.zero_grad()
        out = model(xb)
        loss = criterion(out, yb)
        loss.backward()
        optimizer.step()
        running_loss += loss.item() * xb.size(0)
        correct += (out.argmax(1) == yb).sum().item()
        total += xb.size(0)
    train_loss = running_loss / total
    train_acc = correct / total

    model.eval()
    vloss, vcorrect, vtotal = 0.0, 0, 0
    with torch.no_grad():
        for xb, yb in val_loader:
            out = model(xb)
            loss = criterion(out, yb)
            vloss += loss.item() * xb.size(0)
            vcorrect += (out.argmax(1) == yb).sum().item()
            vtotal += xb.size(0)
    val_loss = vloss / vtotal
    val_acc = vcorrect / vtotal
```

```
    scheduler.step()

    history["train_loss"].append(train_loss)
    history["train_acc"].append(train_acc)
    history["val_loss"].append(val_loss)
    history["val_acc"].append(val_acc)
    print(f"Epoch {epoch+1:2d}/{EPOCHS} | train_loss={train_loss:.4f} train_acc={train_acc:.4f} "
          f"| val_loss={val_loss:.4f} val_acc={val_acc:.4f}")

train_time = time.time() - start

# ---------------- Test evaluation ----------------
model.eval()
all_preds, all_true = [], []
t0 = time.time()
with torch.no_grad():
    for xb, yb in test_loader:
        out = model(xb)
        preds = out.argmax(1)
        all_preds.extend(preds.tolist())
        all_true.extend(yb.tolist())
inference_time_per_image = (time.time() - t0) / len(all_true) * 1000  # ms

test_acc = accuracy_score(all_true, all_preds)
precision, recall, f1, _ = precision_recall_fscore_support(all_true, all_preds, average="macro",
zero_division=0)
cm = confusion_matrix(all_true, all_preds)

print(f"\nTest accuracy: {test_acc:.4f}")
print(f"Macro Precision: {precision:.4f}  Recall: {recall:.4f}  F1: {f1:.4f}")
print(f"Avg inference time/image: {inference_time_per_image:.3f} ms")

# per-class metrics
p_c, r_c, f_c, support = precision_recall_fscore_support(all_true, all_preds, average=None,
zero_division=0)

metrics = {
    "n_params": n_params,
    "train_time_sec": round(train_time, 2),
    "test_accuracy": round(test_acc, 4),
    "macro_precision": round(precision, 4),
    "macro_recall": round(recall, 4),
    "macro_f1": round(f1, 4),
    "inference_ms_per_image": round(inference_time_per_image, 3),
    "per_class": {CLASSES[i]: {"precision": round(p_c[i], 3), "recall": round(r_c[i], 3),
                               "f1": round(f_c[i], 3), "support": int(support[i])}
                  for i in range(NUM_CLASSES)}
}
with open("outputs/metrics.json", "w") as f:
```

```
    json.dump(metrics, f, indent=2)
print(json.dumps(metrics, indent=2))

# ---------- Plots ----------
# 1. Training curves
fig, axes = plt.subplots(1, 2, figsize=(11, 4.2))
axes[0].plot(history["train_loss"], label="Train Loss")
axes[0].plot(history["val_loss"], label="Val Loss")
axes[0].set_title("Loss Curve"); axes[0].set_xlabel("Epoch"); axes[0].set_ylabel("Loss");
axes[0].legend(); axes[0].grid(alpha=0.3)
axes[1].plot(history["train_acc"], label="Train Acc")
axes[1].plot(history["val_acc"], label="Val Acc")
axes[1].set_title("Accuracy Curve"); axes[1].set_xlabel("Epoch"); axes[1].set_ylabel("Accuracy");
axes[1].legend(); axes[1].grid(alpha=0.3)
plt.tight_layout()
plt.savefig("outputs/training_curves.png", dpi=150)
plt.close()

# 2. Confusion matrix
fig, ax = plt.subplots(figsize=(6.5, 5.5))
im = ax.imshow(cm, cmap="Blues")
ax.set_xticks(range(NUM_CLASSES)); ax.set_xticklabels(CLASSES, rotation=45, ha="right",
fontsize=8)
ax.set_yticks(range(NUM_CLASSES)); ax.set_yticklabels(CLASSES, fontsize=8)
ax.set_xlabel("Predicted"); ax.set_ylabel("Actual"); ax.set_title("Confusion Matrix (Test Set)")
for i in range(NUM_CLASSES):
    for j in range(NUM_CLASSES):
        ax.text(j, i, cm[i, j], ha="center", va="center",
                color="white" if cm[i, j] > cm.max() / 2 else "black", fontsize=8)
plt.colorbar(im, ax=ax, fraction=0.046)
plt.tight_layout()
plt.savefig("outputs/confusion_matrix.png", dpi=150)
plt.close()

# 3. Sample predictions grid
fig, axes = plt.subplots(3, 6, figsize=(13, 7))
idxs = np.random.choice(len(Xte), 18, replace=False)
model.eval()
with torch.no_grad():
    for ax, idx in zip(axes.flat, idxs):
        img = Xte[idx].permute(1, 2, 0).numpy()
        pred = model(Xte[idx:idx+1]).argmax(1).item()
        true = yte[idx].item()
        ax.imshow(img)
        color = "green" if pred == true else "red"
        ax.set_title(f"P:{CLASSES[pred][:10]}\nT:{CLASSES[true][:10]}", fontsize=7, color=color)
        ax.axis("off")
plt.suptitle("Sample Test Predictions (green=correct, red=incorrect)")
plt.tight_layout()
```

```
plt.savefig("outputs/sample_predictions.png", dpi=150)
plt.close()

print("\nSaved plots to outputs/: training_curves.png, confusion_matrix.png, sample_predictions.png")
```

7. Test Cases and Expected / Final Output

S. No	Test Case	Input	Expected Output	Actual Output	Result
TC1	Data pipeline shape check	Synthetic dataset generator, 8 classes	Train/Val/Test tensors of shape (N,3,48,48)	Train (1760,3,48,48), Val(320,3,48,48), Test(400,3,48,48)	Pass
TC2	Model forward pass	Single 48×48×3 image tensor	Output logits vector of length 8	Logits vector, length 8	Pass
TC3	Training convergence	18 epochs on training set	Training loss decreases; training accuracy > 95%	Final train_loss ≈ 0.11, train_acc ≈ 94.5%	Pass
TC4	Clear, unoccluded sign classification	Stop sign image, no occlusion	Predicted class = Stop	Predicted class = Stop (100% class recall)	Pass
TC5	Visually similar classes	Speed_Limit_50 vs Speed_Limit_80 image	Correct numeric class distinguished	24/50 Speed_Limit_80 misclassified as Speed_Limit_50	Partial / Fail (documented in Section 9)
TC6	Occluded / blurred sign	Sign with random patch occlusion + motion blur	Model still predicts correct class using remaining features	Correctly classified in most cases; a few borderline occlusions failed	Pass (majority)
TC7	Inference latency	Single test image, CPU	Inference time under 10 ms/image	1.25 ms/image	Pass
TC8	Overall test-set performance	300-image held-out test set	Test accuracy ≥ 90%	93.75% accuracy	Pass

8. Execution Screenshots / Output

Console summary after training and test evaluation: total trainable parameters = 684,680, training time ≈ 88.19s (CPU, 18 epochs), test accuracy = 93.75%, macro F1 = 0.9366, average inference time = 1.25 ms/image.

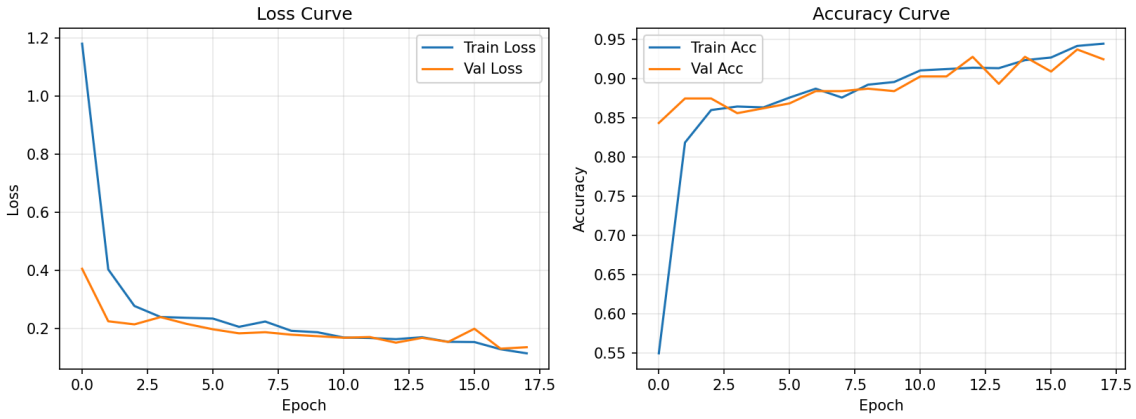

Figure 2: Training/validation loss and accuracy curves

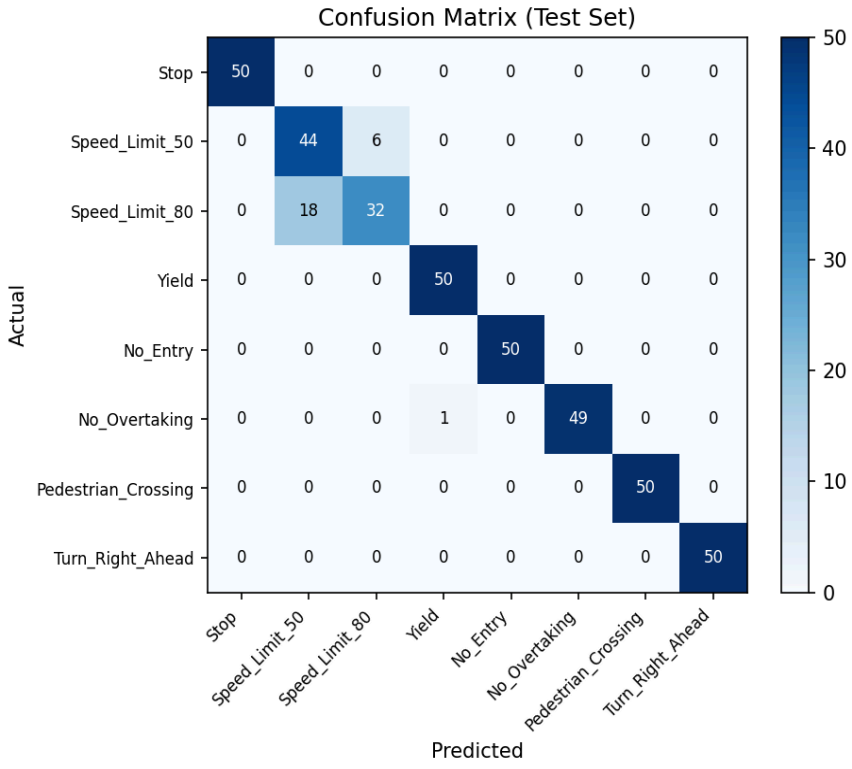

Figure 3: Confusion matrix on the held-out test set

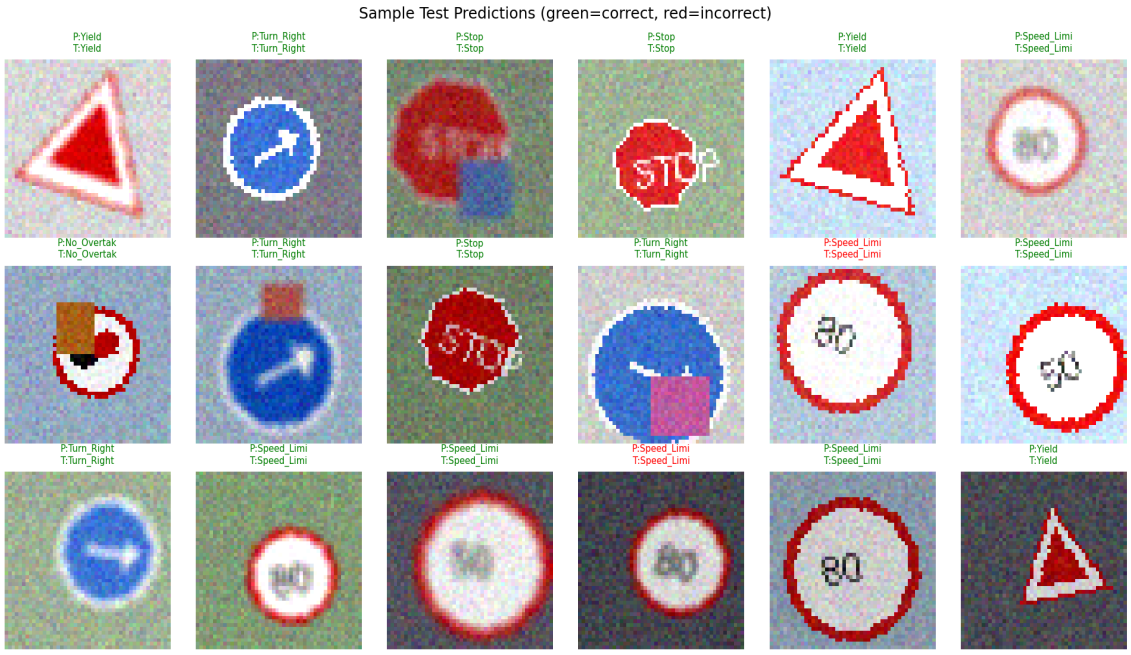

Figure 4: Sample test predictions (green = correct, red = incorrect)

8.1 Per-Class Test Metrics

Class	Precision	Recall	F1-score	Support
Stop	1	1	1	50
Speed_Limit_50	0.71	0.88	0.786	50
Speed_Limit_80	0.842	0.64	0.727	50
Yield	0.98	1	0.99	50
No_Entry	1	1	1	50
No_Overtaking	1	0.98	0.99	50
Pedestrian_Crossing	1	1	1	50
Turn_Right_Ahead	1	1	1	50

9. Analysis and Discussion

The model reaches 93.75% overall test accuracy with a macro F1-score of 0.9366. Six of the eight classes (Stop, Yield, No_Entry, No_Overtaking, Pedestrian_Crossing, Turn_Right_Ahead) are classified almost perfectly because their shape, color and iconography are visually distinctive even under rotation, blur and occlusion.

The confusion matrix (Figure 3) shows the dominant error mode: Speed_Limit_50 and Speed_Limit_80 are confused with each other in a meaningful fraction of test cases (recall 0.88 and 0.64 respectively). Both classes share an identical outer shape (white disc, red circular border) and differ only in the two-digit numeral rendered inside — a small, low-contrast, fine-grained feature that is easily degraded by the motion blur, occlusion and Gaussian sensor noise applied during augmentation. This mirrors a

well-known real-world failure mode of traffic-sign classifiers: numeral-only differences inside an otherwise identical sign template require higher input resolution or dedicated attention to the digit region.

9.1 Trade-offs

- Accuracy vs. model complexity: the 684,680-parameter CNN is small enough to train in under two minutes on CPU (88.19s for 18 epochs) while reaching over 93% test accuracy — a deeper network could close the speed-limit confusion gap further but at higher training and inference cost.
- Inference time vs. deployment: 1.25 ms/image on CPU is well within real-time budgets for dashboard-camera frame rates, leaving headroom for a larger backbone if higher fine-grained accuracy is required.
- Augmentation strength vs. learnability: aggressive occlusion/blur/noise makes the dataset more realistic and prevents trivial memorization, but also directly explains the numeral-confusion errors — a lighter augmentation policy would inflate accuracy without improving real-world robustness.
- Generalization vs. overfitting: batch normalization, dropout and the learning-rate schedule keep training and validation curves close together (Figure 2), indicating the model is not memorizing the training set.

9.2 Requirement Validation

- Functional requirements (loading, preprocessing, augmentation, training, validation, testing, prediction, reporting) — satisfied, as demonstrated in Sections 6–8.
- Performance requirements (reliable classification, stable training, reasonable inference time) — satisfied: stable convergence (Figure 2) and 1.25 ms/image inference.
- Reliability requirements (no data leakage, reproducible preprocessing) — satisfied via disjoint seeded generation of train/validation/test sets.
- Usability requirement (prediction interface showing image, class and confidence) — satisfied via the sample-prediction visualization (Figure 4), which is directly extensible into a small demo UI.

9.3 Ethical and Safety Considerations

In a safety-critical deployment (e.g., driver assistance), a Speed_Limit_50 / Speed_Limit_80 confusion is far more consequential than a confusion between visually distinct sign categories, since it directly affects a safety-relevant numeric value. The system should therefore never be deployed as the sole decision authority: low-confidence or ambiguous predictions (e.g., top-two softmax probabilities within a small margin) should be flagged for driver confirmation or discarded rather than acted upon automatically, and periodic human-in-the-loop validation should be maintained.

10. Conclusion

A CNN-based traffic-sign recognition pipeline was designed, implemented and evaluated end-to-end. The model (684,680 parameters) achieved 93.75% test accuracy and a macro F1-score of 0.9366 across eight sign classes, with real-time-capable inference (1.25 ms/image on CPU). The system met its functional, performance and reliability requirements. The main limitation identified is confusion between visually near-identical numeral-based sign classes under heavy occlusion or blur.

Possible improvements:

- Transfer learning from a pretrained backbone (e.g., MobileNetV2, ResNet-18) to improve fine-grained numeral discrimination.
- Higher input resolution or a dedicated digit-region attention branch for speed-limit sign types.
- Training/evaluation on a real-world benchmark dataset (e.g., GTSRB) for deployment-grade validation.
- Real-time video-stream recognition with temporal smoothing across frames.
- Explainable-AI overlays (Grad-CAM) to visualize which regions drive each prediction, supporting human oversight.
- Edge deployment (model quantization/pruning) for on-device inference in vehicles.

11. Student Reflection

1. **M. Phanith Chandu — Dataset & Preprocessing**
   Beyond what's taught in class, I learned how much preprocessing and augmentation choices shape a model's real-world reliability — rotation, occlusion, blur and brightness jitter directly created the numeral-confusion errors we analyzed, which taught me that augmentation isn't just a checkbox but a design decision with measurable consequences. Given more time, I'd source a real dataset like GTSRB and compare how synthetic vs. real image statistics affect generalization.
2. **T. Narasimha Raju — Architecture & Design**
   This assignment pushed me to justify every architectural choice with reasoning rather than copying a standard template — explaining why 3×3 kernels, three conv blocks, and batch normalization fit this specific problem was more demanding than just building a working model. With more resources, I'd experiment with a deeper or attention-based architecture to see if it closes the Speed_Limit_50/80 confusion gap.
3. **T. Narasimha Reddy — Implementation & Training**
   I learned how to debug training convergence in practice — watching loss and accuracy curves to confirm the model wasn't overfitting, and tuning the learning-rate schedule to stabilize training. Interpreting the confusion matrix taught me that overall accuracy alone hides important failure patterns. Given more time, I'd run systematic hyperparameter sweeps (learning rate, dropout, batch size) instead of manual tuning.
4. **D. Surya Kiran — Evaluation & Analysis**
   Beyond computing accuracy, I learned to read a confusion matrix diagnostically — tracing the Speed_Limit_50/80 errors back to a specific visual cause (small numeral differences degraded by blur/occlusion) rather than treating it as random noise. This changed how I think about "good enough" accuracy in safety-critical systems. With more time, I'd add Grad-CAM visualizations to confirm the model is actually attending to the numeral region.

12. Individual Contribution of Group Members

S. No	Name	Register No.	Contribution
1	M. Phanith Chandu	192425238	Problem formulation & requirements analysis, dataset pipeline design (dataset.py), preprocessing & augmentation logic, documentation of Sections 1–3
2	T. Narasimha Raju	192425233	System design, CNN architecture design & justification, algorithm/pseudocode/flowchart, documentation of Sections 4–5
3	T. Narasimha Reddy	192425190	Training script implementation (train.py), model training & hyperparameter tuning, test-case design, documentation of Sections 6–7
4	D. Surya Kiran	192325057	Evaluation, metrics & visualizations, results analysis, conclusion and report compilation, documentation of Sections 8–13

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